

Suryanarayana P S

Chennai, India (Open to Relocate to Germany)

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Dear Hiring Manager,

I'm applying for Embedded Firmware Engineer (Motor Control) roles in Germany, where you need engineers who can ship reliable real-time firmware for BLDC/PMSM drives and validate it on hardware. Across 5+ years, I have delivered motor control and motor-interface firmware in Embedded C, built deterministic control loops around PWM/ADC timing, and closed field and bench issues through disciplined debugging with oscilloscopes and logic analyzers.

In my current role at Grundfos, I work with legacy pump-control firmware and convert requirements into clean, modular components that remain stable under real constraints. I develop Embedded C modules for control and motor interfaces, add unit tests for critical logic, and use trace hooks to shorten debug loops. I partner closely with hardware and integration teams to drive review-ready fixes and protect release quality.

Previously at TI Clean Mobility, I led motor MCU integration and validation for electric powertrain benches. I executed waveform-based failure analysis, built repeatable test checklists, and aligned firmware and hardware teams on clear acceptance criteria. That structure reduced repeat failures and helped teams converge faster during re-test cycles.

At Vitalcore Technologies, I developed and deployed control features used across fans and HVAC units. I implemented FOC, MTPA, and speed/power controllers, and built calibration routines to handle board-level variation. I also created validation and qualification firmware to automate PCB checks, expand edge-case coverage, and cut test effort. This hands-on mix of control, firmware, and validation fits teams that build inverter-driven products and expect engineers to own behavior end-to-end.

If your team builds motor drives for pumps, appliances, industrial automation, or e-mobility subsystems, I can contribute immediately across firmware delivery, fault handling, and verification. I'm ready to relocate to Germany and work in a product environment that values stable releases, measurable validation, and clean embedded design.

Sincerely,

Suryanarayana P S